

Preparing Students for the AI Era: Integrating Relevant Skills into Modern Education

RESPONSE TO TEXTS

NAME: LIM YU HAN

DATE: 17/05/2025

References	Key Ideas	Critical Analysis	Synthesis/Summary	Comparison/Contrast	Problem&solution
Crompton & Bruke (2023) Title: Artificial intelligence in higher education: the state of the field Source: International Journal of Education Technology in Higher Education	AI use has grown very fast in recent years, especially in 2021 and 2022. Most AI tools are used to help with things like grading, giving feedback, predicting student success, tutoring and organizing learning.	Strengths: -Focus on practical AI applications rather than just theoretical concepts. -Highlights gaps in current research, such as the lack of studies from developing nations. Weakness: Overlooks some critical issues, including the difficulties students and teachers encounter when adopting these technologies.	AI has now been incorporated into university classrooms both as a subject of study and as a tool to improve education. While many institutions use AI to automate processes such as grading, far fewer focus on teaching students how to understand or develop real AI systems. This means that with the incorporation of AI into education, students still do not understand the basics of what it is and how it can be useful in their future careers. Although Crompton and Bruke (2023) do not advocate for the expansion of secondary AI education, their research suggests such a shift in the focus is essential. In short, we	When compared with Walter (2024), both articles discuss the lack of preparation students have for using AI effectively. However, while this article focuses on the absence of AI literacy in general education, Walter(2024) provides more concrete strategies like teaching prompt engineering to build technical AI skills.	A problem that the article underscores is that universities are trying to use AI tools to enhance systems, but they are not teaching students what AI is or how it functions. Without basic AI literacy, students will not be able to get ready for future employment opportunities. Thus, it is essential to introduce AI education earlier in schooling, not only as an elective for computer science major students.

References	Key Ideas	Critical Analysis	Synthesis/Summary	Comparison/Contrast	Problem&solution
		Relevance: Pinpoints the disconnect between AI adoption and student comprehension.	need to stress the integration of AI and engineering education to teach students experiential learning and AI literacy.		
Walter (2024) Title: Embracing the future of Artificial Intelligence in the classroom: the relevance of AI literacy, prompt engineering, and critical thinking in modern education Source: International Journal of Educational Technology in	This article highlights the importance of AI literacy, prompt engineering, and critical thinking as essential skills for students in the AI era. It also shows how students often misuse AI tools like ChatGPT because they don't know how to ask the right questions or evaluate AI responses.	Strengths: -Author combines personal experience with research, making the suggestions more practical. -Article talks about both the benefits and the dangers of using AI Weakness: - Focuses a lot on the Swiss university and may not apply exactly the same in other countries or	Simply integrating AI into schools is not enough. First and foremost, students must be taught how to interact with AI systems in a purposeful and evaluative manner. As Walter (2024) has drawn attention to, AI tools can be helpful, but their value hinges on the students' ability to ask sensible questions and evaluate the answers given. This is pertinent to my area of interest on how to prepare learners for a future with AI technology. Besides, it has been shown that both pupils and teachers have a basic lack of understanding and	Compared to Qian et al. (2025), which focuses on how AI systems support interdisciplinary learning, this article focuses more on developing individual student skills. While both articles emphasize the need for smarter AI use in education, Walter (2024) zooms in on the student experience, and Qian et al. (2025) highlights systemic change.	Many students are using AI applications without understanding their inner workings or how to maximize their efficiency. This reflects an absence of basic skills and critical thinking. The proposed solution is to restructure teaching approaches with clear frameworks for example practical evaluations and instructional scaffolding to better enable the students to master AI (Walter, 2024).

References	Key Ideas	Critical Analysis	Synthesis/Summary	Comparison/Contrast	Problem&solution
Higher Education		schools with fewer resources. Relevance: It connects AI with classroom learning and shows the real need for students to develop skills—not just use AI, but understand and manage it properly.	resources to effectively engage with AI. In general, foundation skills such as prompt design and AI system understanding are vital today, they need to be integrated into teaching and learning frameworks and standards.		
Qian et al. (2025) Title: Influence of artificial intelligence on higher education reform and talent cultivation in the digital intelligence era Source: Scientific Reports	The authors introduce the SEOM model which is an AI-driven system that recommends personalized learning paths. The model recommends paths by analyzing student data and linking concepts across disciplines. Furthermore, this	Strengths: -Focus on solving real problems in education, like poor resource management and disconnected learning systems. -Shows how AI can help students learn in a more personalized and efficient way. it's doing.	The article broadens my perspective about AI's purpose in higher education by showing how it can transform learning methods at their core rather than only providing instructional support. Besides, the article also show that educators from different academic specialties can create personalized learning pathways using systems such as SEOM. AI technology now proves applicable in multiple	Compared to Karataş et al. (2024), which focuses on how teachers use AI to change classroom content, this article is more focused on AI systems that analyze data to guide learning decisions. Both support curriculum change, but through different methods:	Traditional teaching methods often treat subjects separately and don't connect learning across different fields. The SEOM model solves this by using AI to analyze student behavior, link learning topics together, and suggest personalized learning paths across different disciplines.

References	Key Ideas	Critical Analysis	Synthesis/Summary	Comparison/Contrast	Problem&solution
	system helps move away from rigid teaching methods toward more flexible and adaptive learning experiences.	Weakness: Paper is very technical. Relevance: As reported by (Qian et al., 2025), AI can improve education systems in measurable ways.	academic fields beyond computer science as this discovery validates its importance in interdisciplinary education. In addition, Qian et al. (2025) demonstrates that AI-enhanced systems can transform educational frameworks into more adaptive and intelligent platforms.	teacher-driven versus system-driven.	
Karataş et al. (2024) Title: Reshaping curriculum adaptation in the age of artificial intelligence: Mapping teachers' AI-driven curriculum adaptation patterns Source: British Educational	The article explores how teachers adapt their classroom materials using AI tools like ChatGPT. It reveals that while teachers are actively reshaping content, many do so without clear guidelines or understanding of the ethical risks involved.	Strenghts: -The article combines both numbers (from surveys) and stories (from interviews), which gives a more complete picture. -It shows how teachers from different backgrounds use AI in unique ways, depending on their	This article elucidates AI's dual role in education. The technology aids student learning while giving teachers the power to completely revise their teaching methods. To illustrate, educators now have the ability to deliver personalized lessons and create dynamic materials while removing outdated content through ChatGPT applications. The trend of teachers interacting with AI as active experimenters who	Compared to Kurtz et al. (2024), which offers institutional-level strategies for AI integration, this article focuses more narrowly on individual teacher practices. Both address the need for support, but Kurtz et al. (2024) is broader in scope while Karataş et al. (2024) is classroom-specific.	There is a lack of ethical training and awareness when teachers use AI tools to change lesson content. The study warns that many of them are using AI without proper guidelines or knowledge of its risks. The solution suggested by Karataş et al. (2024) is to give teachers more support through professional development.

References	Key Ideas	Critical Analysis	Synthesis/Summary	Comparison/Contrast	Problem&solution
Research Journal		subject and experience. Weakness: It doesn't go deep into ethical issues. Relevance: Gives us a good idea of how AI can change curriculum design and the teacher's role.	modify these technologies to better meet student needs stands out as especially significant. As shown above, AI transforms educational delivery systems beyond student learning at the instructional level.		
Kurtz et al. (2024) Title: Strategies for Integrating Generative AI into Higher Education: Navigating Challenges and Leveraging Opportunities	This article discusses how generative AI (GenAI) is impacting higher education. And then it presents a four-level model for integrating AI into the classroom. Importantly, the article highlights both opportunities	Strengths: Clearly lays out the risks and gives real advice for what schools should do next. Weakness: Talks a lot about institutional planning and less about individual student experience.	The article examines AI's revolutionary influence on higher education beyond a simple teaching aid to demonstrate its disruptive impact on pedagogical methods and cognitive structures. Also, the analysis demonstrates that technical knowledge alone is insufficient to prepare for an AI-integrated future. Without a doubt, educators and	Compared to Crompton & Bruke (2023), which focuses on current AI use in classrooms, this article focuses on future planning and institutional strategies. Crompton & Bruke (2023) describes what's happening now, while Kurtz et al.	The main problems discussed in this article is that both teachers and students are struggling to keep up with the fast changes brought by GenAI. It is obviously that education systems will fall behind if they don't act quickly. In order to solve this problem, the article proposes a four-phase model of AI adoption and highlights the need to

References	Key Ideas	Critical Analysis	Synthesis/Summary	Comparison/Contrast	Problem&solution
Source: Education Sciences	and risks of technology (Kurtz et al., 2024).	Relevance: Connects deeply to student learning by emphasizing the need for digital skills, ethics, and flexible teaching methods.	administrators need to develop flexible perspectives to make the most of AI integration in higher education.	(2024) outlines what needs to happen next.	train everyone in adaptability and lifelong learning (Kurtz et al., 2024).

APPENDIX

2C LIM YU HAN.pdf

ORIGINALITY REPORT

0%
SIMILARITY INDEX

0%
INTERNET SOURCES

0%
PUBLICATIONS

0%
STUDENT PAPERS

PRIMARY SOURCES

Exclude quotes
Exclude bibliography

On
On

Exclude matches < 5%

